

Supplementary Information: Interformer: An Interaction-Aware Model for Protein-Ligand Docking and Affinity Prediction

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1 Supplementary Methods: Training related

1.1 Training data

We trained an energy model using the PDBBind 2020 general set, the published date of PDB is before the 2020 year, and adhering to the same split used by DiffDock. We included 2,599 peptide-like small molecule samples and excluded samples that could not pass RDKit sanitization and PDB exists in CASF2016 test set in order to avoid data leaking. We have 15,686 samples for training and 945 for validation, used for checkpoint selection. After training the energy model, we utilized it to perform re-docking on the entirety of the PDBBind 2020 general set to generate additional docking poses. For each sample, 20 docking poses were generated, resulting in 392,406 samples. Of these, 87,579 were positive samples ($\text{RMSD} < 2 \text{ \AA}$), and 304,827 were negative samples ($\text{RMSD} > 2 \text{ \AA}$), which were used for the training of the affinity and pose score models.

1.2 Preparation

We adhered to the preparation procedure outlined in the study of Posebusters [1]. The conformation of the ligand for docking is determined by selecting the minimum energy conformation from 30 conformations generated using the Universal Force Field (UFF) [2]. The reference ligand and the UFF ligand are added with hydrogen atoms by Open Babel [3]. We employ the reduce program for the protein preparation, enable the correct addition of hydrogen atoms, and provide correct protonation information.

1.3 Training summary

All models were trained using eight A100 GPUs, and the training process could be completed within a day. We employed a standard warm-up learning rate scheme for training. To improve generalization for unseen proteins, we implemented a balanced sampling strategy to alleviate the Out of Distribution (OOD) problem, as detailed in section 1.6. An energy model was trained for docking using standard hyper-parameter settings, as listed in Table S1. Although we believe that the predictive performance of the energy functions could be augmented through additional energy model ensembles, we reserve this exploration for future work. For a comparison with the GNINA method, which uses an ensemble of five model predictions, we trained four models for pose and affinity scoring. These models were trained with random seeds and a random selection of typical learning hyper-parameters, as presented in Table S1. To efficiently learn the inter interactions of inter blocks in the affinity model, we applied an extra mask to the attention calculation from Eq.6 when the distance of protein-ligand atom pairs exceeded $5.3 \text{ \AA}$.

1.4 Ablation study

In this section, we seek to find out the importance of each component in the Interformer architecture, and we will attempt to either remove or replace each primary component with a standard module to compare the performance of the baseline model with the

	Energy Model	Pose-Affinity Model
Num of GPUs (A100, 32GB)	8	8
Num of Models	1	4
Batch Size	24	80
Peak-Learning Rate	0.0012	0.0003/0.0004/0.0007/0.0006
Warmup Steps	11000	10000/10000/20000/16000
Dropout Rate	0.1	0.1/0.1/0.1/0.1
Attention Dropout Rate	0.1	0.1/0.2/0.2/0.1
Hidden Dimension	128	128
FeedFowardLayer Hidden Dimension	512	512
Number of Intra Blocks	6	6
Number of Inter Blocks	3	3
Number of Weights	3,621,726	3,554,966
Protein-Ligand Maximum Attention Distance	None	5.3 Å
Pocket Cutting Distance	7 Å	7 Å
Consider Cofactor	Yes	Yes
Stop Epoch	85	58/39/50/46
Training Hours	10.97	26.04 /26.06/32.78/28.032

Table S1 The hyper-parameters utilized in the Interformer for energy mode and four pose score and affinity predictions models. Each pose score and affinity model operates within a small range of random values for parameters such as the dropout rate, learning rate, and random seed.

modified ones. The hyperparameters are identical across all models, as described in Table S1 Energy Model. The results are shown in Table S2. The baseline outperforms all others. The labels ‘No Intra’ and ‘No Inter’ indicate the removal of the Intra-Blocks module and the Inter-Blocks module, respectively. ‘No Edge Output Layer’ refers to the direct use of the updated edge feature from the Inter-Blocks, and ‘No Attention Mask’ implies no restrictions due to any attention mask, the model similar to the Graphormer[4] architecture.

1.5 Input embedding

We employ one-hot pharmacophore features to represent the node features of proteins and small molecules, including cofactors within the proteins. Water molecules are temporarily excluded from our system. These pharmacophore features were initially proposed by X-Score [5], comprising 23 dimensions, as shown in Table S3. Additionally, the first edge feature is the Euclidean distance between two atoms. When the complex binding pose is available, the second edge feature is one-hot specific non-covalent interaction between two atoms, a total of 6 dimensions as shown in Table S4.

Model	Posebuster v1	Posebuster v2	PDBBind-TimeSplit
	%RMSD<2Å [†]	%RMSD<2Å [†]	Min Valid Loss [‡]
BASLINE (ENERGY)	78.73	84.09	0.0191
NO INTRA BLOCKS	71.26	77.92	0.0255
NO INTER BLOCKS	75.00	81.81	0.0211
NO EDGE OUTPUT LAYER	75.46	81.16	0.0200
NO ATTENTION MASK	74.76	79.54	0.0206

Table S2 Ablation results on Posebusters benchmark (v1, n=428, v2=308) and PDBBind validation set (n=968), primary components are turned off from the baseline architecture to evaluate the performance of docking accuracy and energy function training loss of validation set. Hyper-parameters used for all models are identical. Bold-formatted numbers represent the best-performing result on each performance metric.

Pharmacophore Atom Type
AliphaticCarbonXSHydrophobe
AliphaticCarbonXSNonHydrophobe
AromaticCarbonXSHydrophobe
AromaticCarbonXSNonHydrophobe
Bromine
Calcium
Chlorine
Fluorine
Iodine
Iron
Magnesium
Nitrogen
NitrogenXSAcceptor
NitrogenXSDonor
NitrogenXSDonorAcceptor
Oxygen
OxygenXSAcceptor
OxygenXSDonorAcceptor
Phosphorus
Sulfur
SulfurAcceptor
Zinc
IsProtein

Table S3 The node features employed in Interformer consist of 22 pharmacophore atom types, along with an additional feature indicating whether the atom belongs to the protein or the ligand.

1.6 Batch balance sampling

Due to the long-tailed distribution of the dataset in the problem under consideration, specific proteins may have over four hundred PDBs (e.g., Carbonic Anhydrase2), shown in Fig. S1. In contrast, others may have only a few or just one PDB. Additionally, negative samples are considered, leading us to modify the original sampling

Type	Description
Distance	Euclidean distance between two atoms.
None	There is no explicit interaction between two atoms.
HydrogenBond	Whenever two centers of hydrogen accetor and hydrogen donnor come with a distance of HYDROGEN_DIST_MAX.
HalogenBond	Halogen bond is similar to Hydrogen bond but for one of the side is halogen element.
SaltBridge	Whenever two centers of opposite charges come with a distance of SALT-BRIDGE_DIST_MAX.
PiStack	π -stacking for two aromatic rings is reported whenever their centers are within a distance of PISTACK_DIST_MAX, the angle deviates no more than PISTACK_ANG_DEV from the optimal angle of 90 for T-stacking or 180 for P-stacking
PiCation	π -Cation interactions are reported for each pairing of a positive charge and an aromatic ring if the distance between the charge center and the aromatic ring center is less than PICATION_DIST_MAX and the offset between the ring center and the charge is no larger than PISTACK_OFFSET_MAX.
Hydrophobic	Hydrophobic interactions are non-polar atoms they are close together.

Table S4 The edge features utilized in Interformer comprise the Euclidean distance and six specific interactions.

strategy as outlined in Algorithm S1. In the proposed approach, we aim to optimize the model by including one protein and all of its ligands in a single batch, which helps balance the weight of each protein in a long-tailed distribution. The sampling of negative samples is restricted to having an equal number of positive samples in one batch and the same ligand.

Algorithm S1 Batch Balance Strategy

- 1: **Input:** Pairs Training Data $\mathcal{D} = \{\langle \mathbf{x}^n, \mathbf{y}^n \rangle\}_{n=1}^N$; Interformer θ ;
 - 2: $T =$ Shuffle targets based on uniprot id
 - 3: $Z = []$
 - 4: **for** t in T **do**
 - 5: $x_{pos} \leftarrow$ Select all positive samples from t
 - 6: $x_{neg} \leftarrow$ Retrieve the same amount of x_{pos} negatives samples with the same ligand from t
 - 7: $Z.append(x_{pos} + x_{neg})$
 - 8: **end for**
 - 9: Split Z by batch_size into B
 - 10: Training by MiniBatch SGD method
 - 11: **for** b in B **do**
 - 12: Update θ with batch loss $\mathbb{E}_{\langle \mathbf{x}, \mathbf{y} \rangle \sim b}$
 - 13: **end for**
 - 14: **return** θ
-

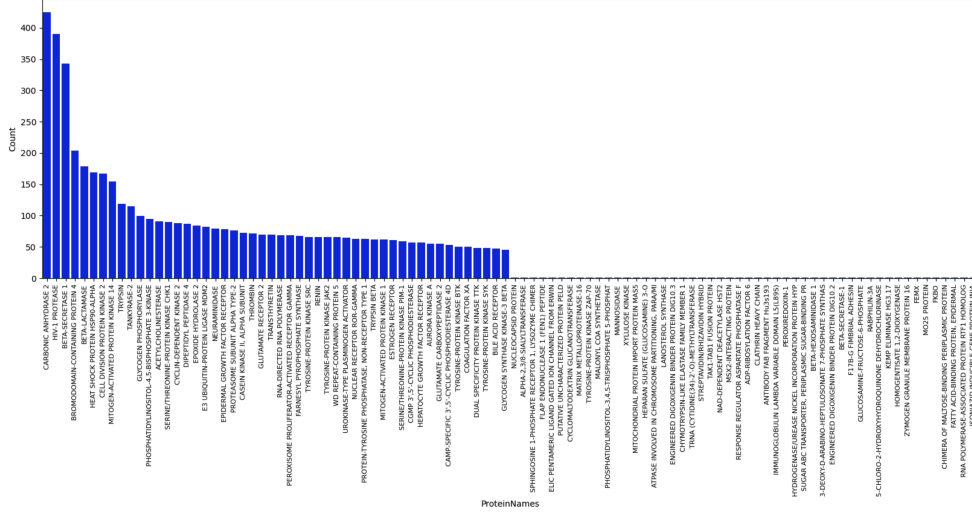


Fig. S1 This figure illustrates the distribution of PDB entries for each protein, sorted in descending order, with a focus on the top 50 and the bottom 50 proteins. Total number of proteins: 4242

2 Supplementary Methods: Inference Related

2.1 Docking

Initially, we employ the Universal Force Field (UFF) [2] to generate the minimized energy conformation of the ligand from its SMILES or SDF (Structure-Data File) representation. The minimized pose and the protein structure are input into the energy model to predict the inter-energy function P for each pair of protein-ligand atoms. P is subsequently combined with the repulsive term of Van der Walls force S to formulate an energy function E as follows:

$$E = E_{intra}(\mathcal{V}_{\mathcal{L}}) + E_{inter}(\mathcal{V}_{\mathcal{L}}, \mathcal{V}_{\mathcal{P}}) \quad (1)$$

$$E_{intra}(\mathcal{V}_{\mathcal{L}}) = \alpha \sum_{i \in \mathcal{V}_{\mathcal{L}}} \sum_{k \in \mathcal{V}_{\mathcal{L}}} S(d_{ik}), \text{ where } i \neq k \quad (2)$$

$$E_{inter}(\mathcal{V}_{\mathcal{L}}, \mathcal{V}_{\mathcal{P}}) = - \sum_{i \in \mathcal{V}_{\mathcal{L}}} \sum_{j \in \mathcal{V}_{\mathcal{P}}} P(d_{ij} | \mathbf{H}_{ij}^a) + \beta S(d_{ij}) \quad (3)$$

$$S(d) = \begin{cases} d^2 & d < 0 \\ 0 & d > 0 \end{cases} \quad (4)$$

where S serves as a penalty term that computes the intersecting VdW radius distance d of two atoms, aiming to prevent steric clashes, the energy function between the ligand atom and the protein atom is represented by E_{inter} , with β acting as the weight of inter-repulsive function. On the other hand, E_{intra} denotes the energy

Method	CASF-2016	
	Pearson	Spearman
Traditional Docking Method		
AUTODOCK VINA	0.604	0.528
GLIDE SP	0.513	0.419
Machine Learning Method		
$\Delta vinaRF_{20}$	0.816	0.750
X-SCORE	0.631	0.604
Deep Learning Method		
GNINA	0.790	-
DEEPDOCK	0.525	0.495
RTMScore	0.507	0.563
PGNN*	0.749	0.748
Our Interformer-Affinity		
ENERGY †	0.468	0.485
AFFINITY w/o NEG †	0.798	0.786
AFFINITY w/ NEG †	0.809	0.802
ENERGY ‡	0.454	0.448
AFFINITY w/o NEG ‡	0.795	0.779
AFFINITY w/ NEG ‡	0.810	0.799

Table S5 Performances of scoring functions on CASF-2016 coreset (n=285), in terms of Pearson correlation coefficient and Spearman ranking coefficient. † means using the crystal structure. ‡ means using the docking pose generated by Interformer. * method is re-performed from this study, and other results are referenced from their respective works. Bold-formatted numbers represent the best-performing result on each performance metric.

function between the ligand atom and another ligand atom, with α denoting the weight of the intra-repulsive function. Lastly, E is the sum of E_{inter} and E_{intra} , is applied as the energy function in MC sampling algorithm section 2.2 to sample candidates poses. We selected the values of α and β to be 30 and 5 through a straightforward, simple, manual parameter selection process.

2.2 Monte Carlo Sampling

We apply the Interformer to predict the energy functions E of each protein-ligand atom pair to generate candidate binding poses. Monte Carlo (MC) sampling, a global optimization procedure, minimizes the energy of the conformer while preserving its chemical plausibility. To ensure this plausibility, the movement of the ligand must be accounted for in the space of $6 + K$ variables, where three are for translation, another three for rotation, and K for the torsion angles of rotatable bonds.

In our implementation, the sampling method is a variant of the AutoDock Vina procedure applied in our MC Sampling. This process unfolds in four stages. Initially, the $6 + K$ variables of the ligand conformation undergo randomization until all ligand atoms reside within the protein pocket auto box (a $20\text{\AA}^3$ box centered on the reference

Type	Description
AutoBoxSize	20 Å * 20 Å * 20 Å
AutoBoxAdd	4 Å
GridResolution	0.125 Å
Exhaustiveness	64
StepsOfEachMC	2000

Table S6 Monte Carlo sampling parameters. The movement space of the ligand is set by the larger value between AutoBoxSize, a cubic space centered on the small molecule, and AutoBoxAdd an additional 4Å to the extreme coordinates of the ligand. GridResolution specifies the finest grid unit during sampling. Exhaustiveness delineates the count of Monte Carlo chains, while StepsOfEachMC signifies the optimization iterations per Monte Carlo chain.

ligand or a box 4 Å extend the maximum or minimum coordinates of the reference ligand). Following this, an iterative optimization of the $6 + K$ variables is executed. Each iteration comprises a ligand conformation mutation and a subsequent BFGS local gradient descent minimization based on energy function E , typically iterating 2000 times per MC chain. The ligand conformation mutation involves one of three randomly selected operations: translating the entire ligand, rotating the whole ligand, or adjusting one of the K torsion angles.

In the third stage, all energy-minimized conformers from all iterations are gathered, clustered, and sorted by energy function E , resulting in selecting the top 20 candidate poses. Each MC chain can run independently in parallel, usually 64. In the final stage, we perform ten steps of post-minimization on the candidate’s poses to alleviate the issue of the steric clash between ligand and protein. This approach enhances the weight of the inter steric clash term in the E during the minimization process. The parameters of the MC sampling are presented in Table S6.

Contributions of MC sampling

We re-implemented the Monte Carlo sampling in C++, distinguishing our method from AutoDock Vina, particularly in the computation of BFGS gradients. Instead of employing an approximation approach for differentiation, our process ensures precise derivation, accelerating the docking speed up to 200 times compared to the PyTorch version. We have also modified the original grid map function module to operate at atom level rather than atom type level, enabling it to adapt to the Interformer energy function or other potential functions like distance map. These modifications have led to significant improvements in both docking accuracy and speed.

3 Supplementary Notes

3.1 Future Works

Several ideas require further exploration to enhance the performance of our model. Here, we outline some directions for future work and potential improvements:

Inclusion of more interaction types: Our model accounts for hydrogen bonds and hydrophobic interactions yet omits other significant interactions such as halogen bonds, ionic bonds, and Pi-Pi stacking. We could integrate additional atom-pair-specific Gaussian functions into our modeling framework to encompass these additional interaction types.

Refinement of energy function: In our approach, individual interactions were modeled using a singular Gaussian function. Future efforts could be directed toward crafting distinct probability functions for various interactions, thereby achieving a more precise characterization of the energy landscape.

Water model: Although our model does not explicitly consider the impact of water molecules, it can accommodate the presence of water within binding sites by integrating water molecules from crystal structures that are positioned with relative accuracy. This method would facilitate the sampling of interactions that involve water bridges, potentially enhancing the docking process.

Data augmentation with limited data: We refine the affinity model by incorporating conformations produced in a single iteration by the energy model. However, the scope of data generated by the model remains limited. Developing a computationally efficient sampling method, such as diffusion-based techniques, could provide a richer dataset, potentially enhancing the model’s accuracy during training.

Other molecules interactions: We focus exclusively on protein-ligand interactions. However, non-covalent interactions are fundamental to all molecular interactions, encompassing protein-protein and protein-nucleic acid associations. Expanding our dataset to include additional crystallographic data from the RCSB database represents a promising avenue for future research.

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